SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology | Department of Agricultural Engineering

AEN 214: INTRODUCTION TO ELECTRONICS

OFFICIAL COURSE SYLLABUS & LEARNING CONTRACT

Instructor: Dr. Kadeghe Goodluck Fue, PhD, P.Eng (T) | Laboratory: Electronics & Precision Agriculture Lab
Course Status: Elective | Credit Rating: 6 Credits | Total Notional Hours: 60 Hours | Academic Year: 2025/2026

1. Course Aim & Rationale

This course serves as a rigorous, hands-on introduction to the fundamental principles, physics, and applications of analog and digital electronics within the domain of agricultural and biosystems engineering. Modern agricultural transformation—ranging from smart drip irrigation automation and climate-controlled greenhouses to precision livestock telemetry, autonomous tractors, and grain quality monitoring—relies on solid-state transducers, discrete analog amplifiers, power semiconductor switching actuators, and digital processing logic. AEN 214 bridges basic electro-technics with modern AgriTech cyber-physical hardware systems.

2. Course Expected Learning Outcomes (CELOs)

By the completion of this course, engineering students will be able to:

1. Describe and evaluate fundamental concepts of analog and digital electronics and articulate their strategic importance in modern agricultural mechanization and automation.
2. Analyze electronic components and determine their electrical characteristics, including linear/non-linear resistors, capacitors, inductors, diodes, and transistors under DC, transient, and high-frequency conditions.
3. Analyze, simulate, and design basic analog circuits—including multi-stage amplifiers, active/passive filters, DC voltage regulators, and timing oscillators—tailored for agricultural sensor conditioning.
4. Synthesize and minimize digital logic systems, utilizing Boolean algebraic laws, De Morgan's theorems, and Karnaugh maps to implement combinational and sequential logic for farm process automation.
5. Explain principles of data conversion, analyzing Analog-to-Digital Converters (ADCs) and Digital-to-Analog Converters (DACs) interfacing physical farm sensors with digital processors.
6. Apply microcontrollers and power semiconductors (Power BJTs, MOSFETs) to drive inductive farm actuators (solenoid valves, motor drivers, relays, and injection injectors).
7. Demonstrate hands-on laboratory mastery in circuit prototyping, breadboarding, schematic design, multimeter and oscilloscope diagnostics, and fault troubleshooting.
8. Evaluate emerging electronics technologies in agricultural engineering, including IoT sensor nodes, solar photovoltaic balance-of-system electronics, and autonomous agricultural robotics.

3. Detailed Course Content & Lecture Modules

Module #	Curriculum Topic	Key Subtopics & Engineering Concepts	Hours Allocated
Lecture 1	Passive Electronic Components	Resistors (types, color codes, power rating, TCR, thermistors, LDRs, strain gauges); Capacitors (electrostatics, dielectrics, codes, transients); Inductors; Reactance (Xc, Xl) and resonance.	6.0 Hours
Lecture 2	Semiconductor Physics	Bohr atomic model; Energy bands (VB, CB, Eg); Intrinsic vs extrinsic; Doping (P-type, N-type); Carrier transport; PN junction depletion layer; Built-in barrier V0; Hall effect sensor operation.	6.0 Hours
Lecture 3	Semiconductor Diode & Circuits	Shockley equation; Dynamic resistance rd derivation; Piecewise approximations; Half-wave and full-wave bridge rectifiers; Smoothing filters and ripple factor; Diode clippers and clampers.	7.0 Hours
Lecture 4	Special Purpose Diodes	Zener diodes (Zener vs Avalanche breakdown); Shunt voltage regulator design; LEDs (direct bandgap, electroluminescence, PAR grow lighting); Schottky barrier diodes (solar bypass, fast recovery); Photodiodes.	6.0 Hours
Lecture 5	Bipolar Junction Transistor (BJT)	BJT physics (NPN, PNP); Biasing rules; CB, CE, CC configurations; Current gains (α, β); DC load line and Q-point; Thermal runaway; Low-side saturated switch; CE small-signal amplifier.	8.0 Hours
Lecture 6	Field Effect Transistor (FET)	JFET operating principles; Pinch-off VP; Shockley transfer equation; Transconductance gm; DE-MOSFET and E-MOSFET; Power MOSFETs (IRF540N) in solar pump and motor drivers; ESD handling.	7.0 Hours
Lecture 7	Number Systems & Arithmetic	Positional radices (Binary, Octal, Decimal, Hexadecimal); Inter-base conversion algorithms; Signed two's complement arithmetic and overflow; Digital codes (BCD, Excess-3, Gray code); ASCII & NMEA.	6.0 Hours
Lecture 8	Introduction to Digital Electronics	Logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR); Universal gate synthesis; Propagation delays; Half and Full adders; Combinational greenhouse interlocks; Latches and Flip-Flops (SR, D, JK).	7.0 Hours
Lecture 9	Boolean Algebra & K-Maps	Boolean axioms and algebraic laws; De Morgan's theorems; Canonical SOP (minterms) and POS (maxterms); 2, 3, and 4-variable Karnaugh Maps (Gray code order); Don't Care (X) optimization; Grain dryer case study.	7.0 Hours

4. Teaching and Learning Methodology

i. Interactive Lectures & Discussions: Interactive lectures covering passive components, semiconductor physics, and digital logic with live mathematical derivations and circuit simulations.
ii. Hands-on Laboratory Experiments: Practical bench sessions in the Electronics and Precision Agriculture Lab: breadboard circuit assembly, oscilloscope waveform measurement, rectifier diode testing, BJT/MOSFET switching, and digital logic gate verification.
iii. Guest Lectures & Industry Insights: Guest lectures from agricultural automation and sensor systems specialists covering tractor electronic control units (ECUs) and solar irrigation installations.
iv. Group Capstone Project & Presentations: Multi-disciplinary student groups design, prototype, and demonstrate a complete functional electronic circuit solving an authentic agricultural challenge (e.g., smart greenhouse misting controller, automated poultry brooder temperature regulator, grain level monitor).
v. Self-Directed Problem Sets: Three tiered open assignments reinforcing analytical calculations, component selection, and safety derating.

5. Assessment Methods & Grading Policy

Assessment Component	Scope / Topic Coverage	Weighting (%)	Format / Deliverable
Continuous Assessment Tests (CATs)	CAT 1 (Topics 1-4); CAT 2 (Topics 5-9)	20%	Closed-book written engineering examinations
Practical Laboratory Reports	Experiments 1 through 6 (Laboratory logbook)	15%	Formal bench reports, circuit schematics, and oscilloscope plots
Open Coursework Assignments	Assignment 1 (Topics 1-3), Assignment 2 (Topics 4-6), Assignment 3 (Topics 7-9)	10%	Take-home comprehensive problem sets (3 assignments)
Group Project & Defense	Autonomous agricultural circuit prototype	15%	Working hardware prototype, technical report, and oral presentation
University Final Examination	Comprehensive coverage (Topics 1 through 9)	40%	Formal 3-hour theoretical and design university examination
Total Final Grade	All Course Components	100%	SUA Grading Scale (A: 70%+, B+: 60-69%, B: 50-59%, C: 40-49%)

6. Prescribed Textbooks & Reading List

1. Boylestad, R. L., & Nashelsky, L. (2018). Electronic Devices and Circuit Theory (11th ed.). Pearson Education.
2. Sedra, A. S., & Smith, K. C. (2020). Microelectronic Circuits (8th ed.). Oxford University Press.
3. Floyd, T. L. (2017). Electronic Devices: Conventional Current Version (10th ed.). Pearson.
4. Mano, M. M., & Ciletti, M. D. (2017). Digital Design: With an Introduction to Verilog HDL (6th ed.). Pearson.
5. Karim, M. A., & Chen, X. (2017). Digital Design: Basic Concepts and Principles. CRC Press.
6. Person (2015). Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering (6th ed.). Pearson. ISBN-10: 9781292076683.

7. Johnson, M. A., & Smith, K. L. (2023). Fundamentals of Electronics: An Introduction. IEEE Transactions on Education, 66(3), 456-465.
8. Brown, R. J., & Davis, P. W. (2023). Basic Electronic Components and Circuits. Journal of Electrical Engineering, 44(2), 78-85.
9. Williams, S. E., & Wilson, T. A. (2023). Introduction to Semiconductor Devices and Circuits. Journal of Applied Physics, 135(6), 065420.